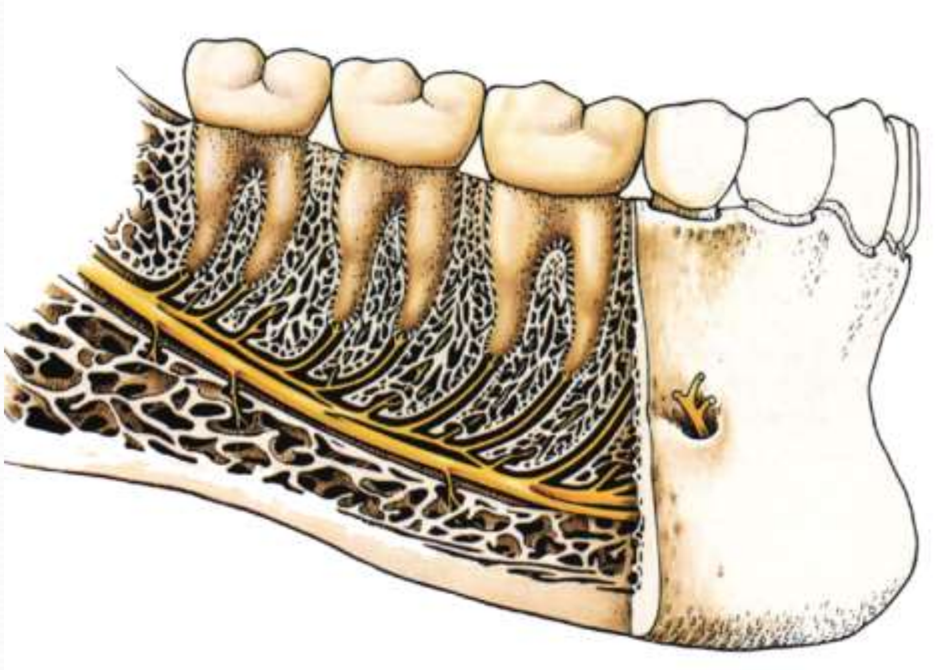
Lecture Mandibular anesthesia

Dr Saravanan R, MDS

3 BDS SEMESTER2

By the end of this lecture you should be able to:

- List the land marks for mandibular anesthetic technique
- Define the different Mandibular Nerve local anesthetic techniques
- Explain the sutability of Mandibular Nerve local anesthetic techniques for mandibular teeth extraction



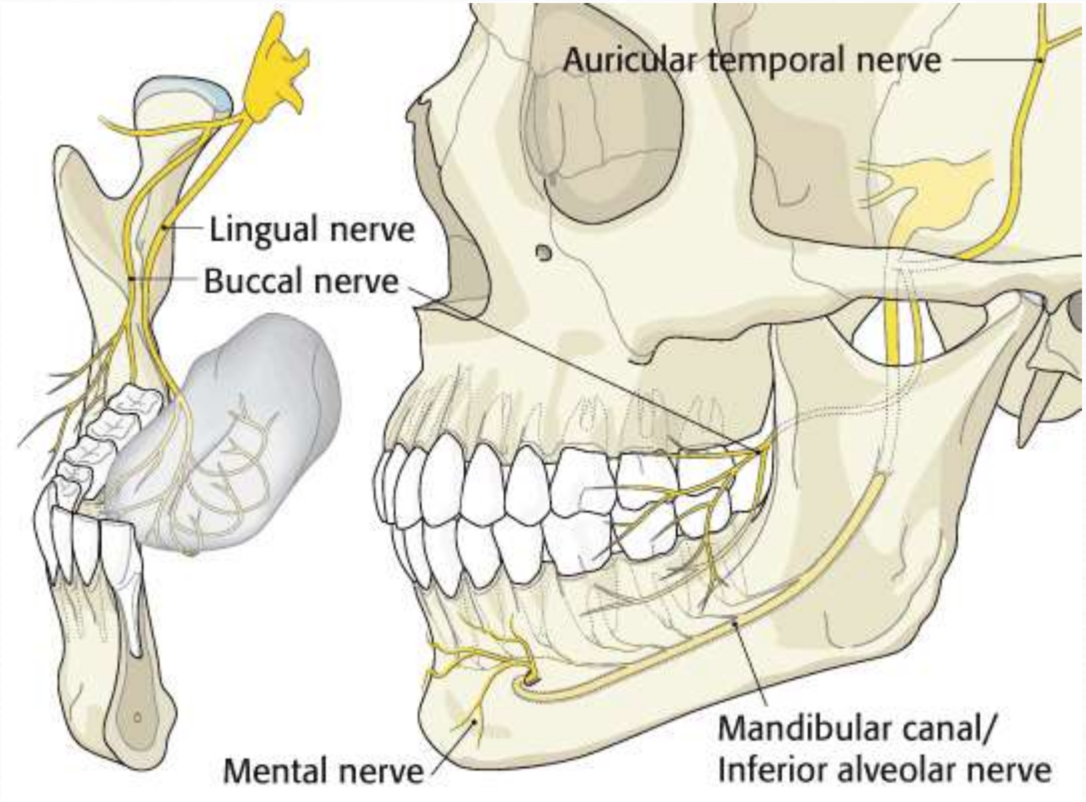
The roots of the lower teeth are embedded in a **thick compact bone**

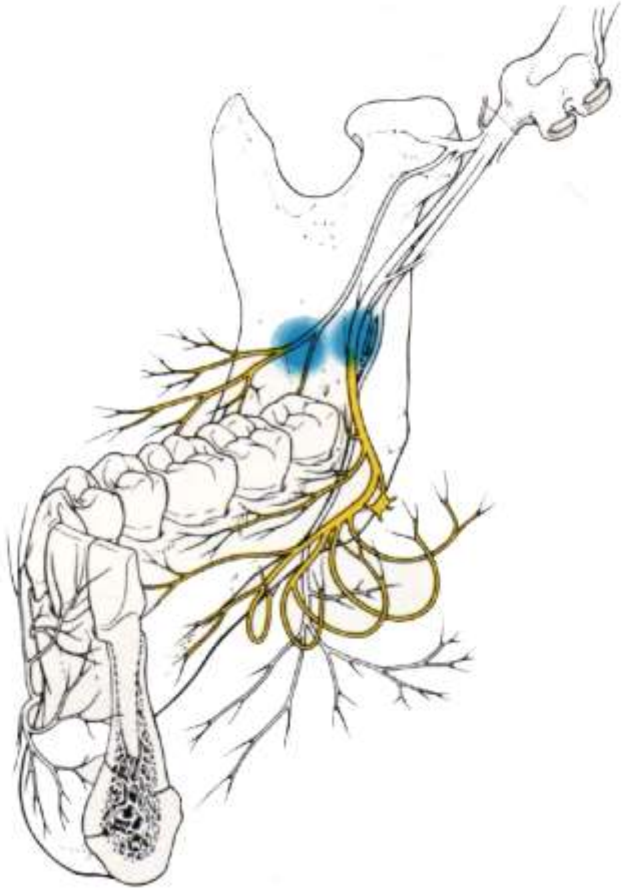
Nerve supply of the mandible

Mandibular teeth are innervated by the **inferior alveolar nerve** which lies in the mandibular canal , The inferior alveolar nerve innervates also the bone of the mandible from the molar area to the midline

•The lingual gingiva and the floor of the mouth is innervated by the **lingual nerve**

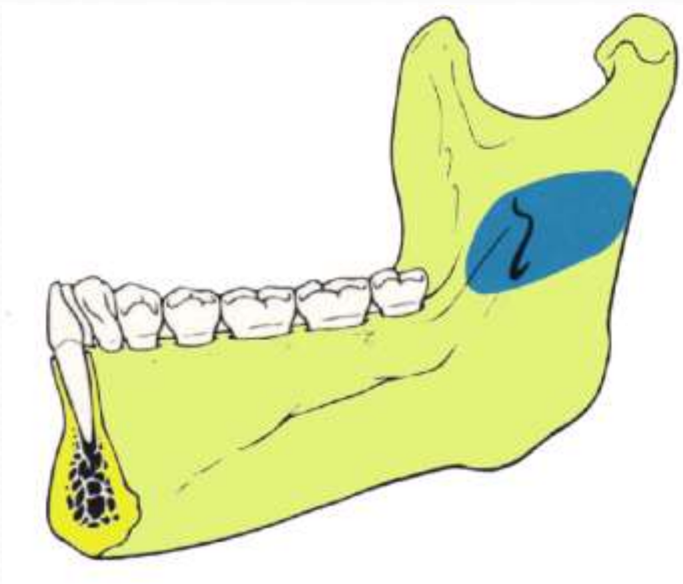
• The terminal branches of the **long buccal nerve** penetrates the buccinator muscle and innervate the buccal gingiva in the lower molar area

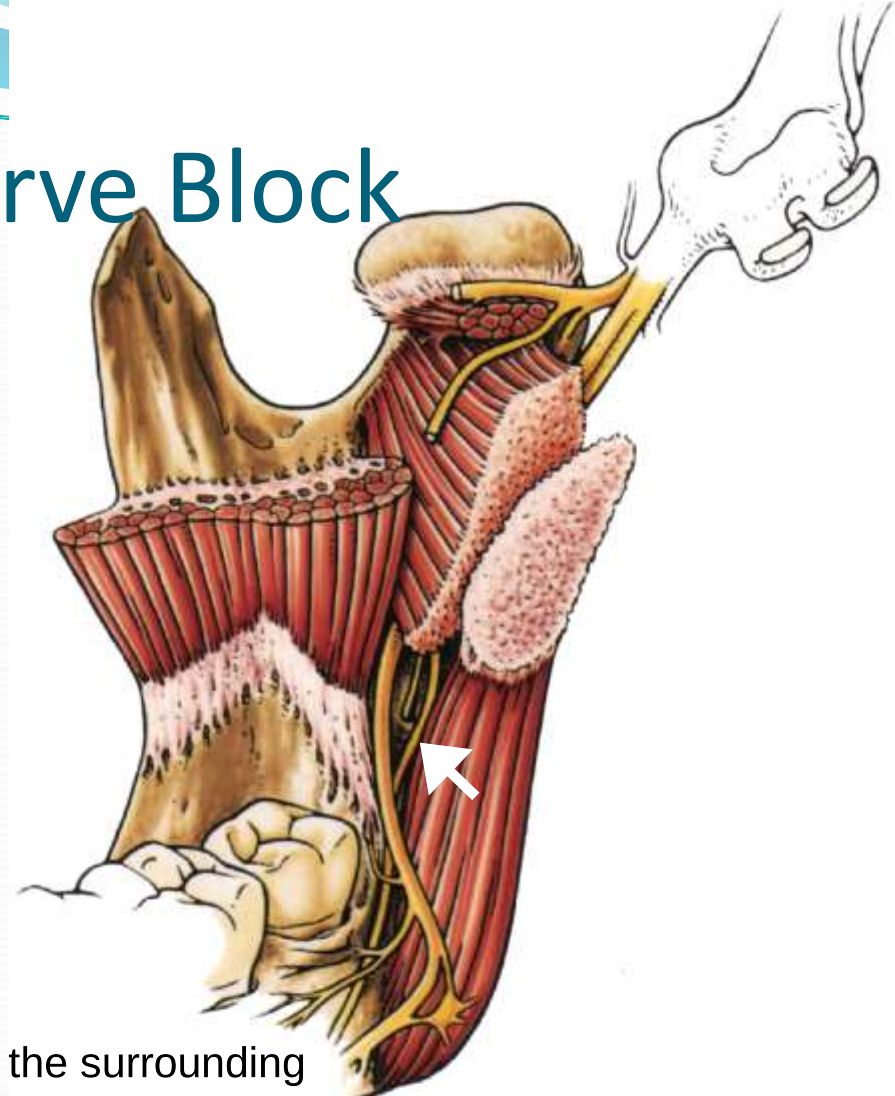




- The Inferior alveolar and lingual nerves can be blocked by one injection at the medial side of ramus
- The long buccal nerve must be blocked by a separate injection on the disto buccal sulcus of third molar tooth

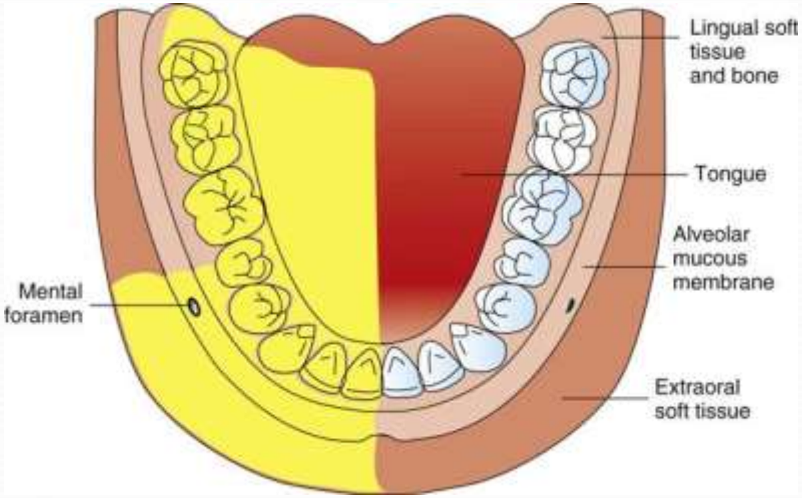
Inferior Alveolar Nerve Block





- The Muscles and soft tissue as well as the surrounding fascia form a space medial to ramus in which lies the inferior alveolar nerve just before it enters the mandibular foramen, the **pterygo-mandiular space**

Areas anesthetized



Central incisor to Third molar teeth pulpal anesthesia

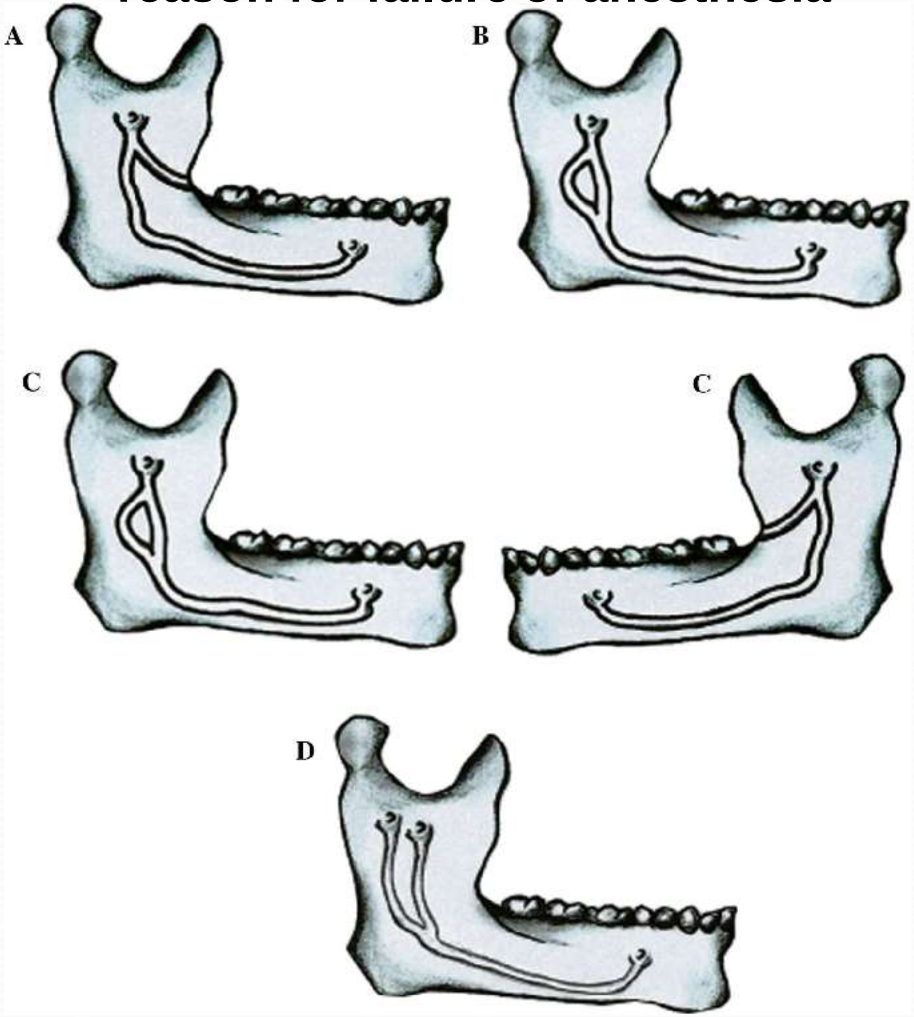
Mandibular body ,inferior ramus

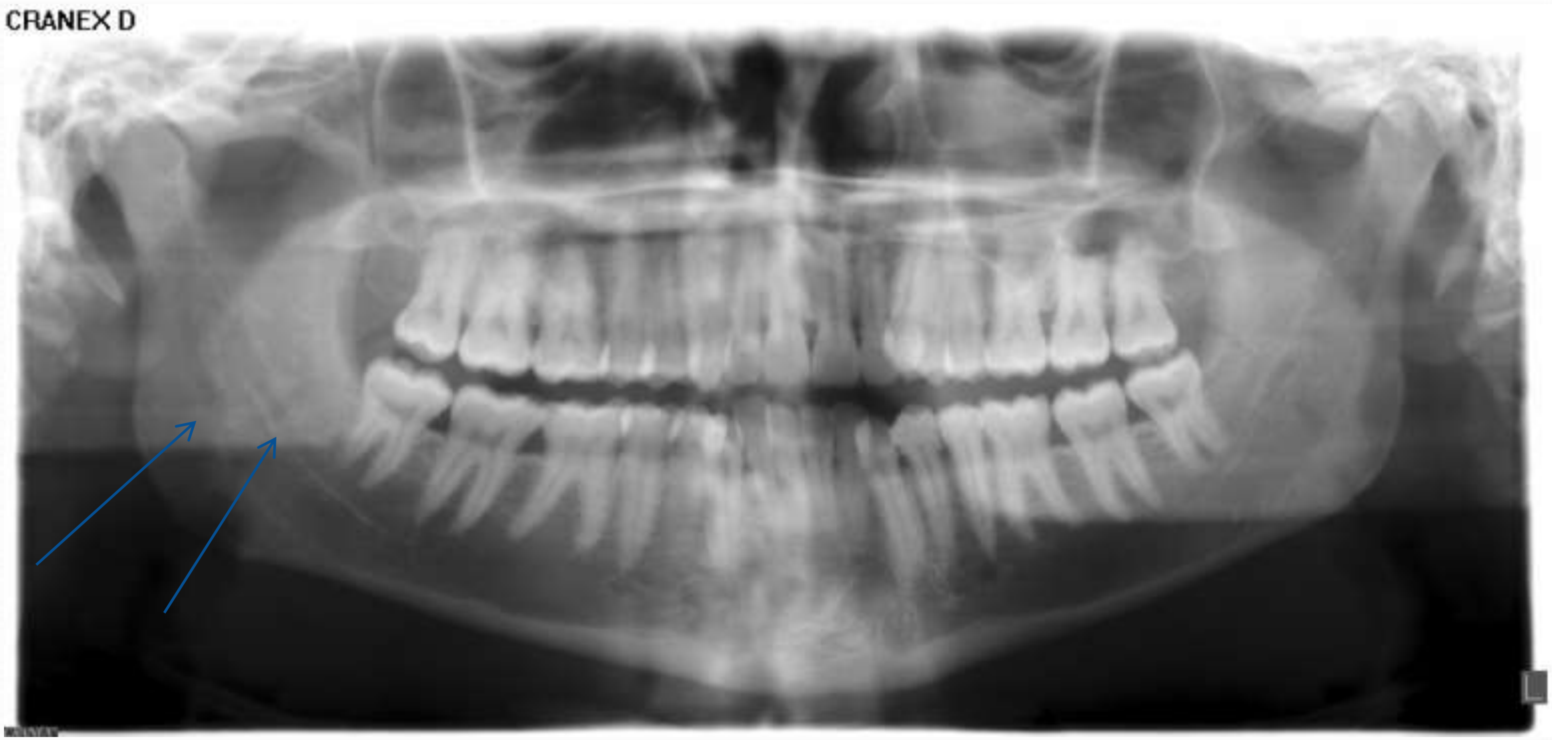
Chin and lip on the same side
Labial gingiva of premolars and incisors

Half the tongue(anterior 2/3) and floor of the mouth, lingual gingiva (by lingual nerve)

- Indications
  - Procedures in multible mandibular teeth
  - When buccal soft tissues anesthesia, anterior to first molar, is required
  - Lingual soft tissue anesthesia
- Contraindication
  - **Infection** in the area of injection
  - Patients who might bite either the lip or the tongue (very young **children** and **mentally handicapped** patients)
- Advantages
  - One injection provide anesthesia for wide area
- Disadvantages
  - **Wide area** of anesthesia (not required for localized procedures)
  - Intraoral **landmarks** not constantly reliable (anatomical variations)
  - Positive aspiration rate (**10%-15%,highest)**
  - Anesthesia Failure rate **15%-20%**
  - **Partial anesthesia when bifid inferior alveolar nerve is present**

bifid inferior alveolar nerve- A reason for failure of anesthesia

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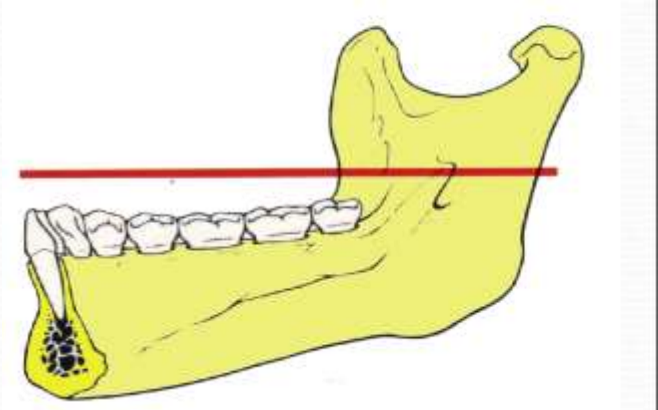
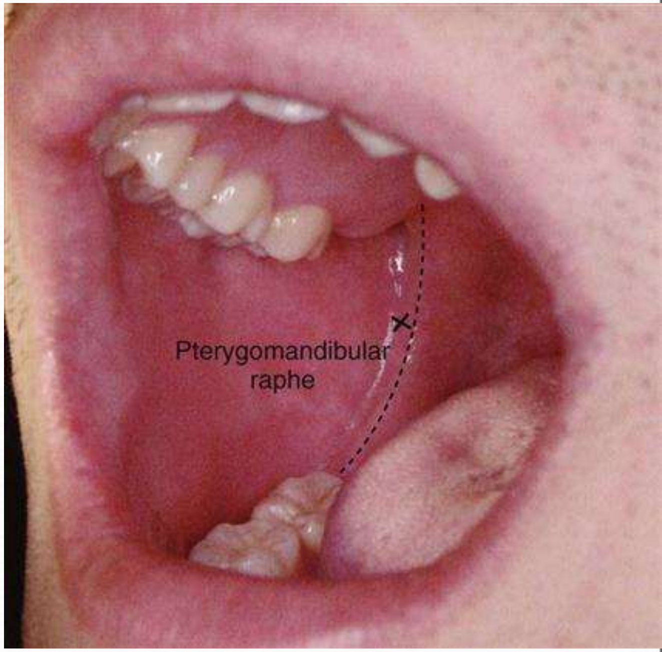
Landmarks





- Coronoid notch, the greatest depression on the anterior border of the ramus
  - Pterygomandibular raphe
  - Occlusal plane of mandibular posterior teeth
- **Orientation of the bevel**
  - Toward bone, but generally not as important as in infiltration techniques
  - Long needle

FIGURE 14-4 Position of the administrator for a (A) right and (B) left inferior alveolar nerve block.

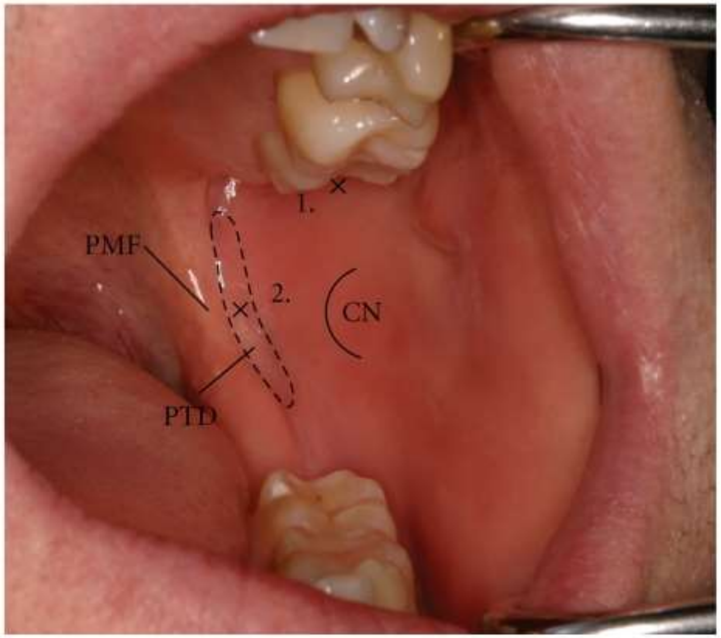
- **Position of the operator**
  - Rt Side: Infront and to the right side of the patient (8 º clock)
  - Lt Side: Behind the patient and to the right side (10 º clock)
- **Position of the patient**
  - Supine or semisupine position

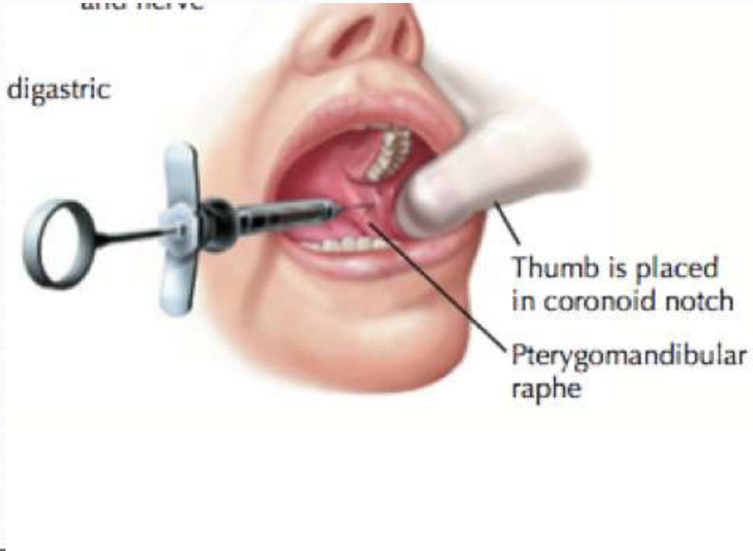
Procedure

Detection of the point of injection

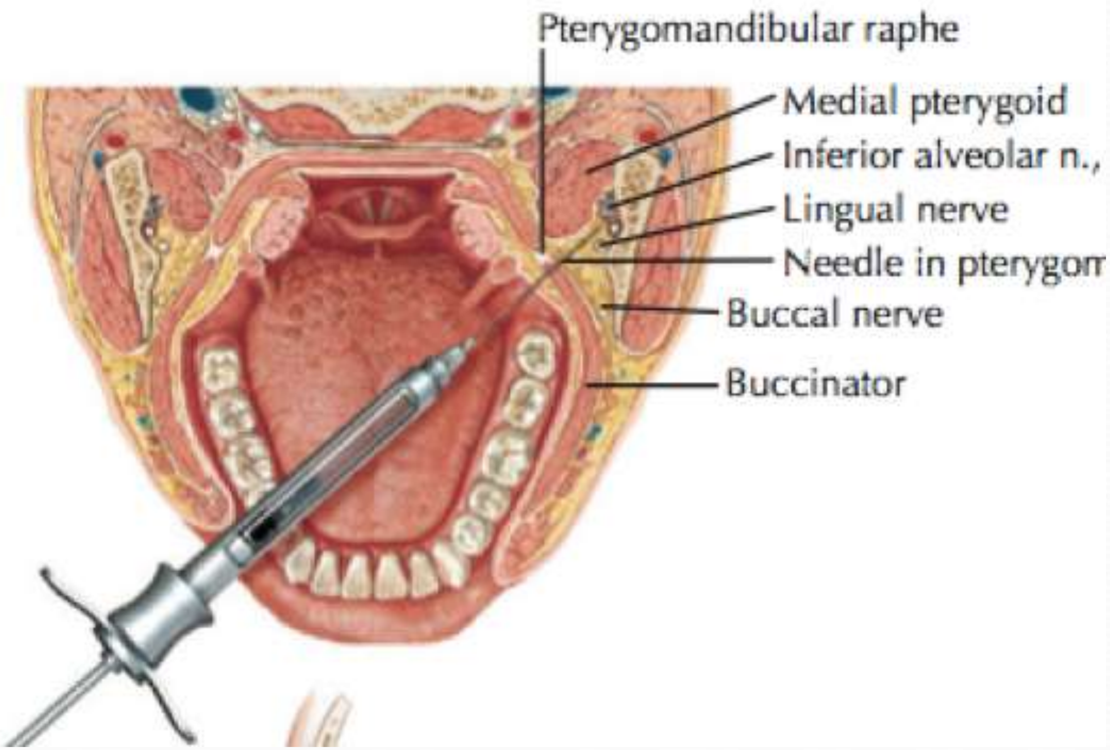
- Index finger of the left hand is placed on the **coronoid notch** (deepest part in the anterior ramus)
- A imaginary line is drawn from the middle of the finger to the **pterygomandibular raphe**

- **Area of insertion**
  - Mucous membrane at the medial aspect of the ramus at point of intersection between two lines:
    - The point of insertion is three fourth of the distance from finger to the pterygomandibular raphe





Insertion of the needle and injection



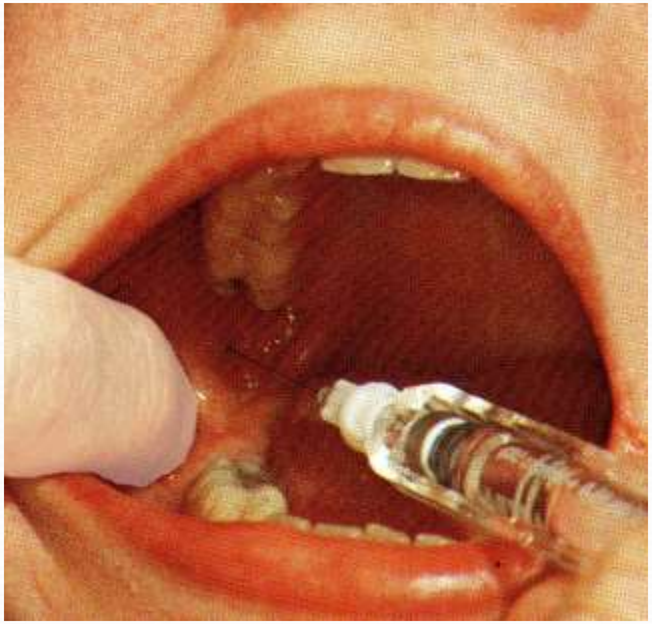
•**Syringe directed** from premolar region of the opposite side at the **level of** the mandibular occlusal plane and slowly advance the needle

•Bone must be contacted at the **depth of 20-25mm**

•When the bone is contacted ,withdraw approximately 1mm and **aspirate**

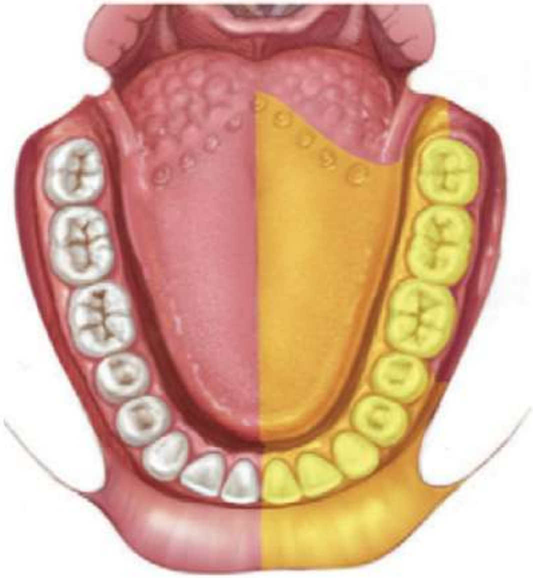
•If negative aspiration slowly **inject 1.5 ml of anesthetic solution** for inferior alveolar nerve

Lingual nerve block



Slowly withdraw the syringe about 10mm and reaspirate and **inject 0.2 ml to anesthetize lingual nerve**

- <u>Signs &symptoms</u>
- Tingling and numbness of lip- indicates anesthesia of mental nerve
- Tingling and numbness of tongue indicates lingual nerve anesthesia
- No pain is felt during procedure



Area anesthetized by an inferior alveolar injection

Failures of Anesthesia

- The most common causes of absent or incomplete IANB follow:
- 1. **Deposition of anesthetic** below the mandibular foramen.
- To correct: Reinject at a higher site (approximately 5 to 10 mm above the previous site).
- 2 .**Deposition of the anesthetic too far anteriorly** on the ramus. To correct: Redirect the needle tip posteriorly.
- **3 .Accessory innervation to the mandibular teeth**
- The primary symptom is incomplete pulpal anesthesia encountered in the mandibular molars (most commonly the mesial portion of the mandibular first molar).
- B .The mylohyoid nerve as the prime reason. (the IAN Block , normally does not block the mylohyoid nerve).

Complications

- **Hematoma (rare)**
- a Swelling of tissues on the medial side of the mandibular ramus after the deposition of anesthetic
- b *Management:* Pressure and cold (e.g., ice) to the area for a minimum of 3 to 5 minutes
- **Trismus**
- Muscle soreness or limited movement
- (1) A slight degree of soreness when opening the mandible is extremely common after IANB (after anesthesia has dissipated).
- Causes of limited mandibular opening after injection is Medial pterygoid muscle injury
- **Transient facial paralysis (facial nerve anesthesia)**
- a Produced by the deposition of local anesthetic into the body of the parotid gland, blocking the VII cranial nerve (facial n.), a motor nerve to the muscles of facial expression. Signs and symptoms include an inability to close the lower eyelid and drooping of the upper lip on the affected side.

Temporary facial paralysis

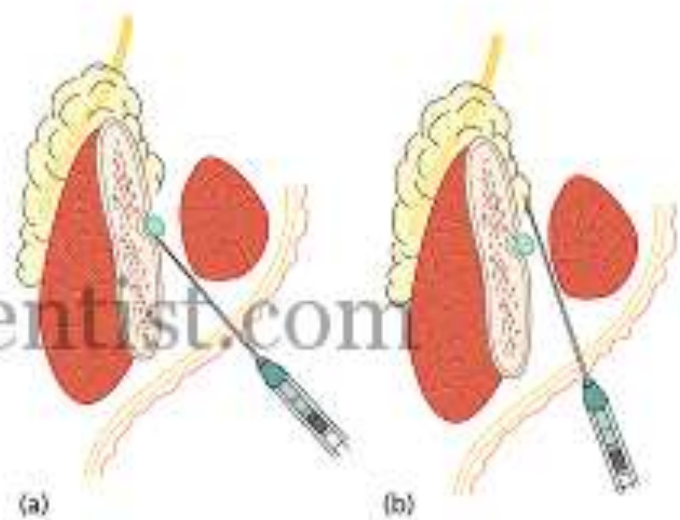
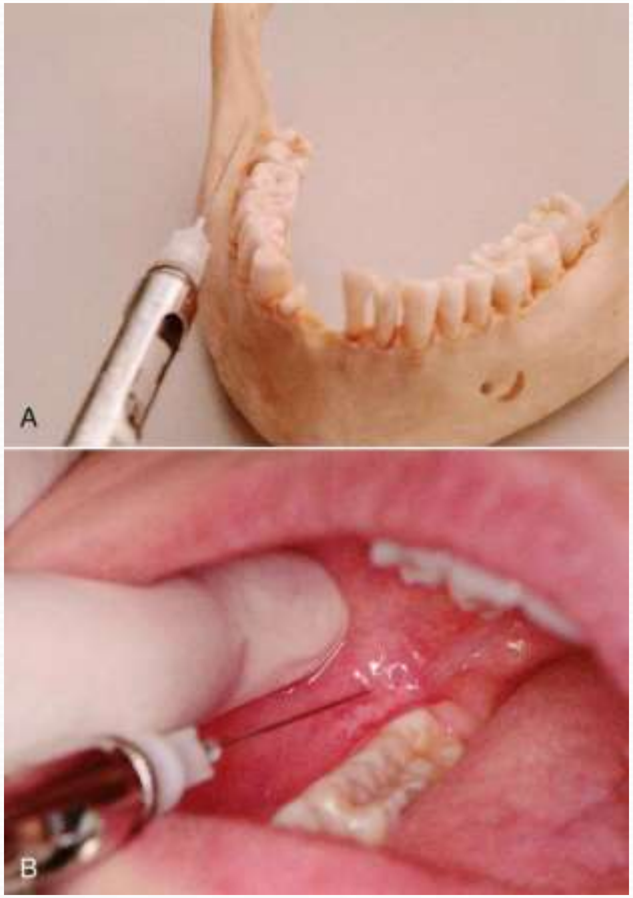


Diagram to show how an inferior alveolar nerve block can affect the facial nerve if the local anesthetic is given too far posterior and the needle does not contact bone: (a) normal, (b) injection into parotid gland where facial nerve can be affected.

Long Buccal Nerve Block

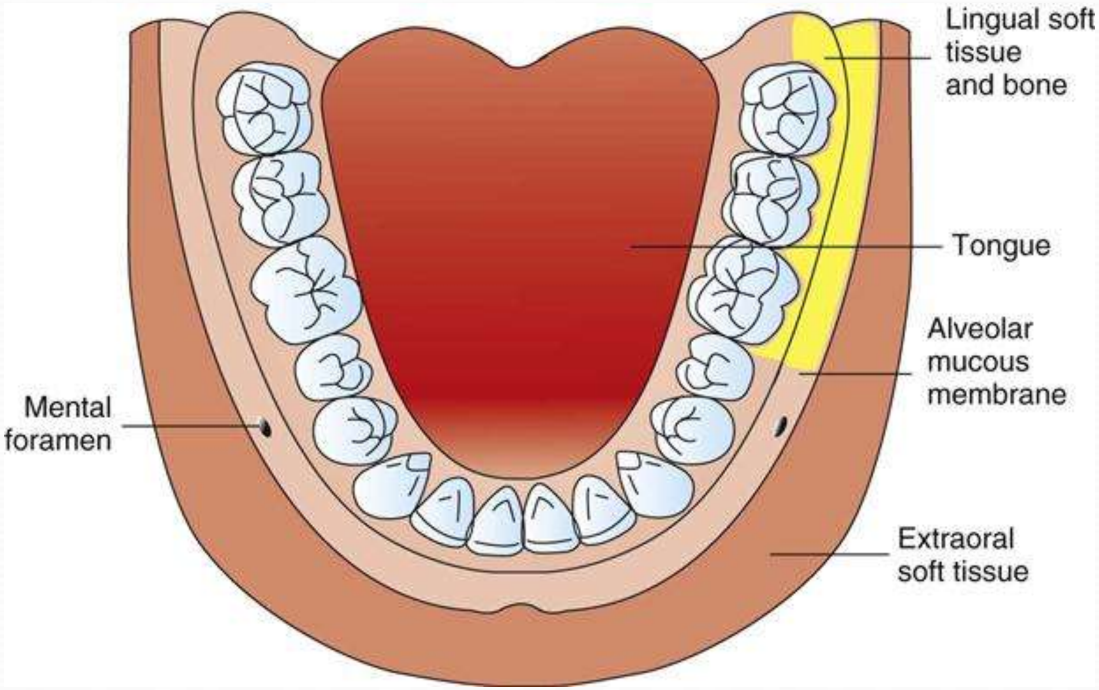


- Long buccal nerve is a **branch of** the anterior division of the mandibular nerve
- **it is not anesthetized in inferior alveolar block**

•Gives innervation to buccal mucosa and gingiva of mandibular molar teeth

•Extractions or any procedures require buccal gingival anesthesia of molar teeth ,long buccal block is needed

Areas of long buccal nerve anesthesia



- **Indications**
- When buccal soft tissue anesthesia is necessary for dental procedures in the mandibular molar region.

Contraindication

- Infection or acute inflammation in the area of injection.

Advantages

- 1 High success rate
- 2 Technically easy

Disadvantages

- Potential for pain if the needle contacts the periosteum during injection.

Positive Aspiration

- 0.7%.

Technique

- 1 A 25- or 27-gauge long needle is recommended. This is most often used because the buccal nerve block is usually administered immediately after an IANB.
- 2 *Area of insertion:* Mucous membrane distal and buccal to the most distal molar tooth in the arch
- 3 *Target area:* Buccal nerve as it passes over the anterior border of the ramus

Injection is made at the point where the nerve crosses the external oblique ridge at the distobuccal aspect of the lower third molar at the level of its occlusal plane

The depth of penetration is usually 1 to 2 mm

Slowly deposit 0.3ml

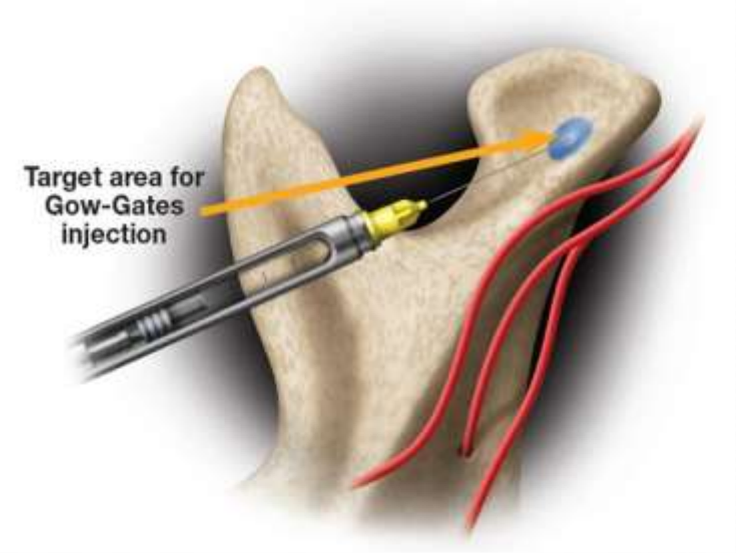
The Gow-Gates Technique:Mandibular nerve block

This technique was originally described in 1973 by George Gaw-Gates, with a high success rate of 90%.

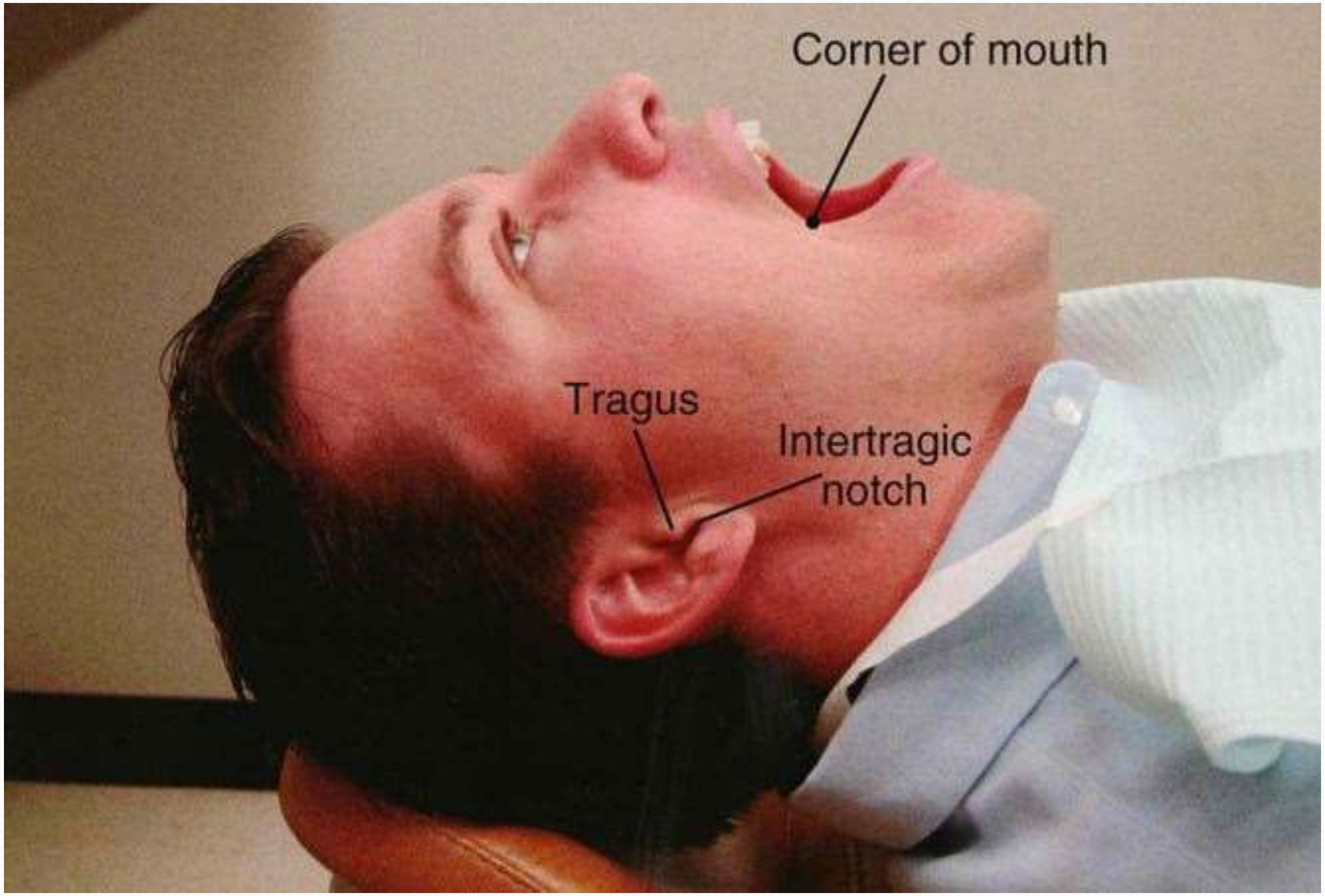
This technique is a true mandibular nerve block since it provides sensory anesthesia to the entire distribution of mandibular division

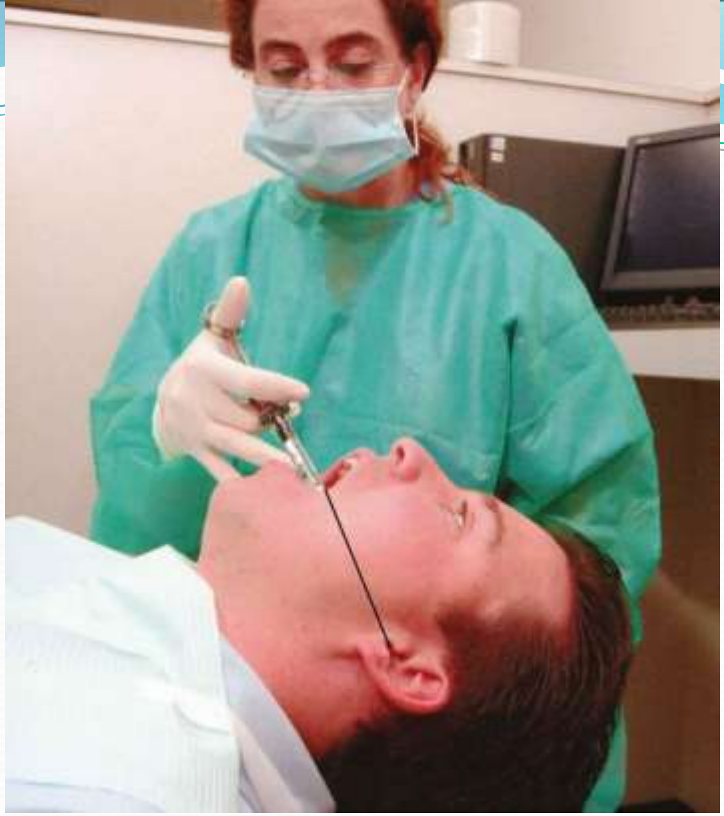
Nerves anesthetized:

1. InferiorAlveolar Nerve.
2. Mental.
3. Incisive.
4. Lingual.
5. Buccal (in 75% of patients).
6. Auroculotemporal.
7. Mylohyoid.



Extraoral landmarks for a Gow-Gates mandibular nerve block





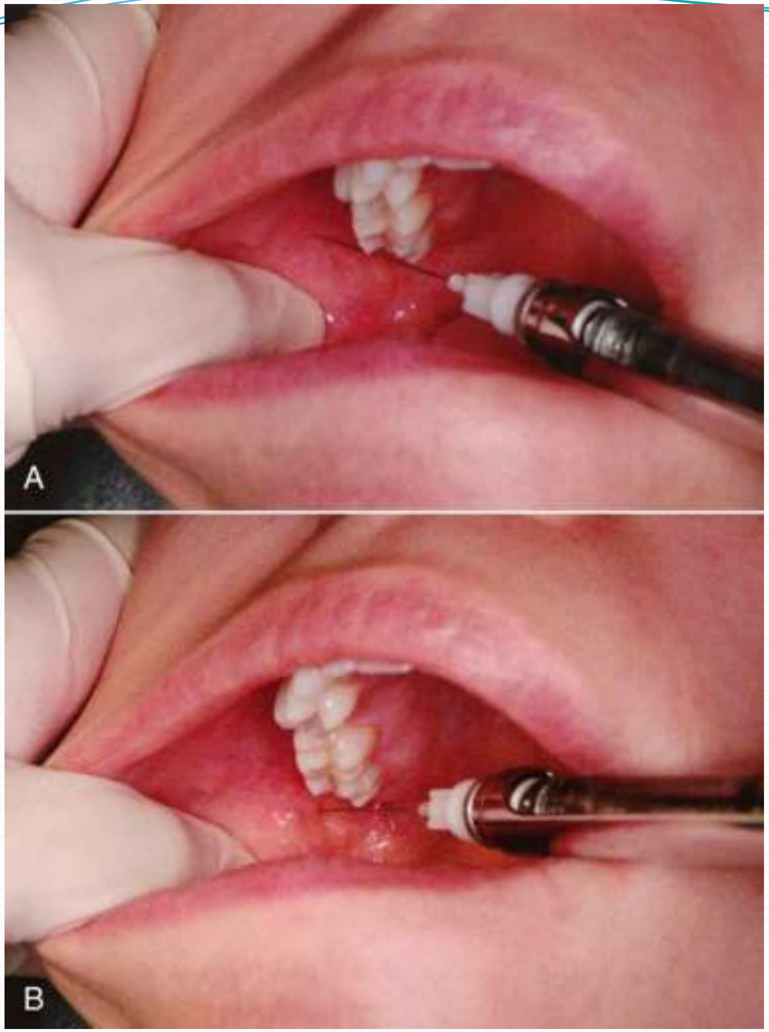
Landmarks:

A. Extra oral:

1. Lower border of the tragus of the ear (intertragic notch).
2. Corner of the mouth.

B. Intraoral:

1. Height of injection established by placement of the needle tip -just below the mesiopalatal cusp of the maxillary second molar.

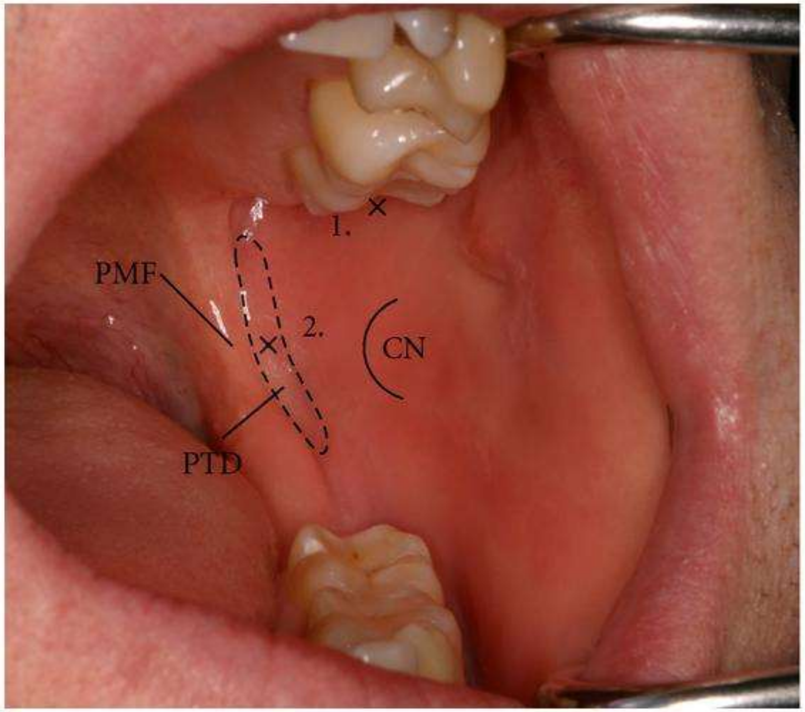


• Align the needle with the plane extending from the corner of the mouth to the intertragic notch ipsilaterally. It should be parallel with the line between the ear and mouth.

• Slowly advance the needle until the condylar neck is contacted. Withdraw the needle slightly and redirect if the bone is not contacted.

• Do not deposit any local anesthetic if bone has not been contacted. The average depth of penetration will be 25 mm .

•Withdraw the needle 1 mm and aspirate, if negative deposit 1.8 ml of LA over 60-90 seconds



1. Site for Gow-Gates technique mandibular block
2. Site for inferior alveolar block

CN-coronoid notch,PMF –pterygomandibular raphe

Akinosi or Closed-Mouth Mandibular Nerve Block

This V3 nerve block was reported originally in 1977 by Joseph Akinosi on a closed mouth patients.

Those patients have limited mouth opening as a result of infection, trauma, post injection trismus.

Nerves anesthetized:

1. Inferior alveolar
2. Incisive
3. Mental
4. Lingual
5. Mylohyoid

technique

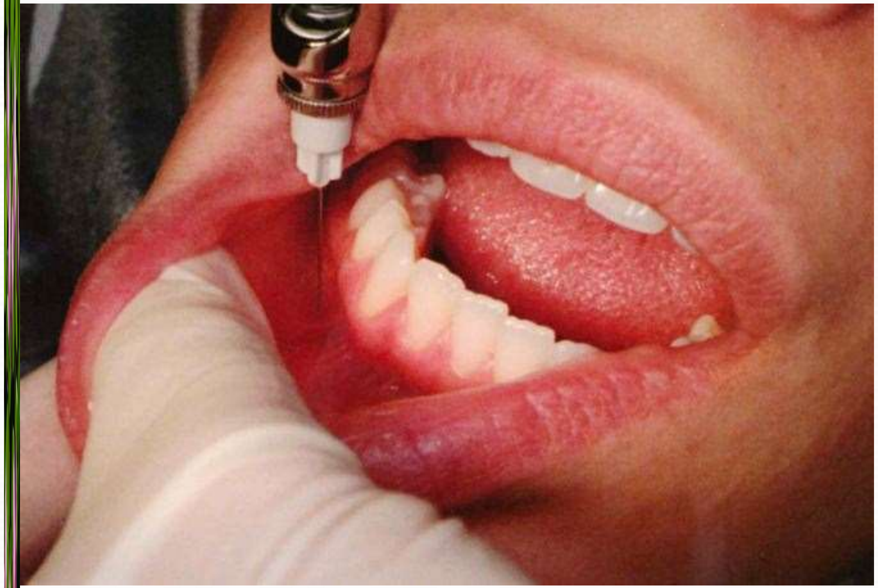
Area of insertion: soft tissue overlying the medial border of the mandibular ramus directly adjacent to the max. tuberosity at the height of the mucogingival junction adjacent to the max. third molar.

Orient the bevel away from the mandibular ramus; thus as the needle advances through tissues, deflection will occur toward the ramus and the needle will remain in proximity to the IAN.

Advance the needle 25 mm into tissues. The tip of the needle should be in the midportion of the pterygomandibular space, close to the branches of V3.

Aspirate, if negative, deposit 1.5 to 1.8 ml of anesthetic solution in 60 seconds.

Mental Nerve Block

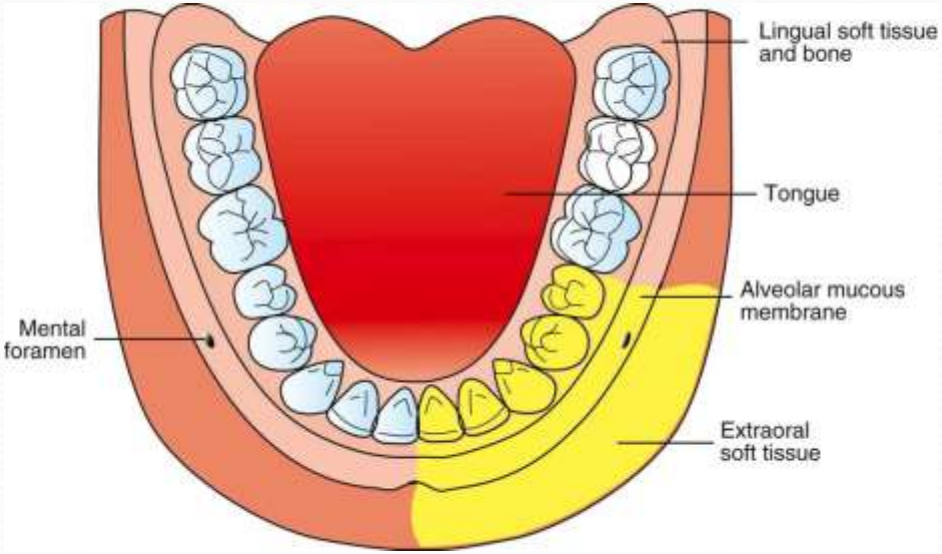


•**Mental nerve exits mental foramen below the apex of the lower premolars**

• A needle oriented towards mental foramen and inserted 5-6 mm depth

• About **0.6 ml** of the solution is deposited slowly when the needle comes in contact with bone

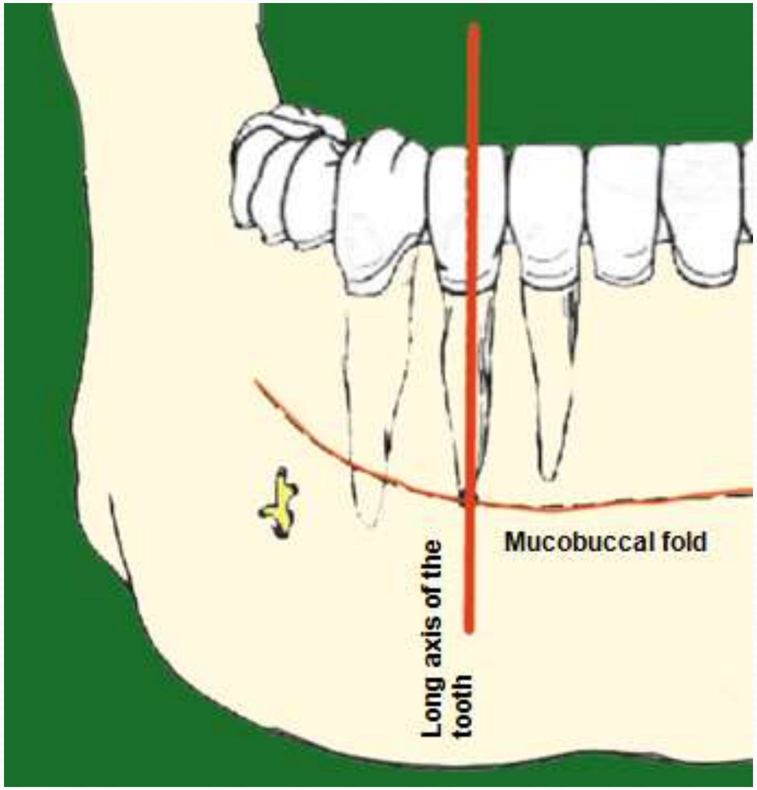
• The area around the foramen is **massaged gently** to push the solution into the foramen (to block the incisive nerve)



Spread of Analgesia after Mental Nerve Block

- The pulp of the premolars and lower incisors
- The labial cortical plate of bone and the peridontium
- The labial mucoperiosteum
- The lower lip and skin over the chin

Labial Infiltration Anaesthesia



Localization of the Point of Needle Insertion

The point of needle insertion is the point of intersection of two imaginary lines.

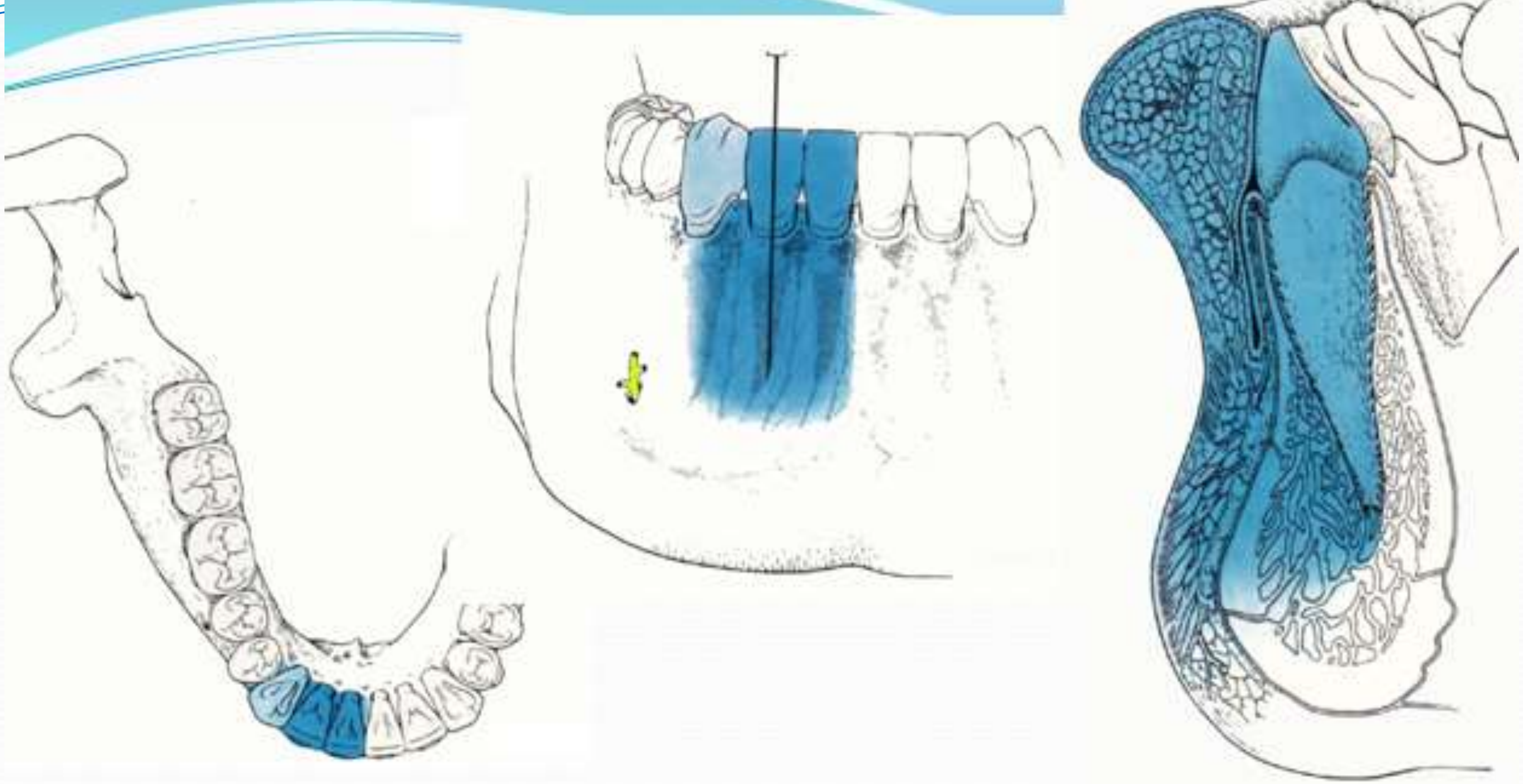
- Horizontal line made by the mucobuccal fold

- Vertical line parallel to the long axis of the tooth and dividing it into two equal halves

Needle Insertion on the Labial Infiltration Injection

45°

- Lift lip and pull tissue taut
- Hold syringe in a line with the long axis of the target tooth and making **45 degree angle** with the buccal plate of bone
- Insert the needle in a steady movement
- Advance the needle until bone is reached
- Aspirate and inject slowly 0.6-0.8 ml, do not permit tissue to balloon
- Slowly withdraw the needle and recap it



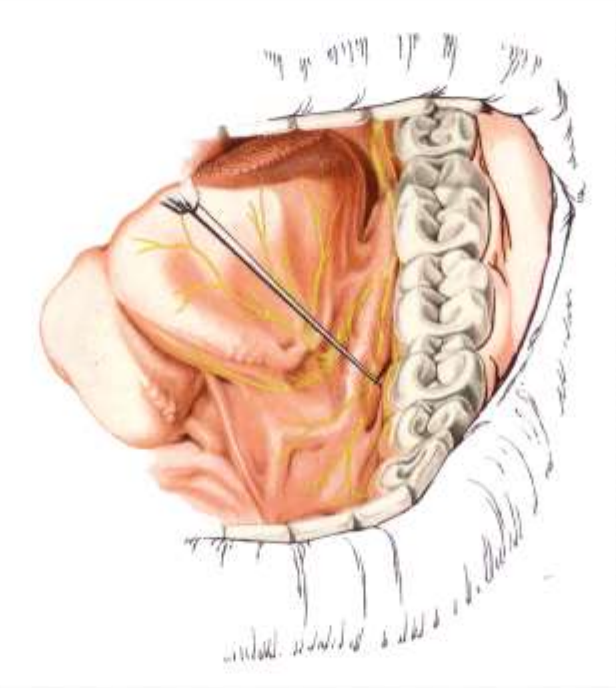
Spread of Analgesia after Labial Infiltration

•The pulp of the target tooth with one or both of the adjacent teeth

• The labial cortical plate of bone and the peridontium

• The labial mucoperiosteum and part of the lower lip

Lingual Infiltration Anaesthesia



- The lingual nerve is anaesthetized by inserting the needle just below the mucosa lingual to the lower incisors or premolars.
- About 0.3-0.5 ml of solution is deposited
- Keep the needle very near to the bone to avoid injuring the blood vessels in the floor of the mouth